

Quick Activity 2

A local coffee shop sells gourmet sandwiches for ₱120 each and brewed coffee for ₱50 a cup. Right now, they're running a promo where you get 15% off, but only on the sandwiches. On top of that, there's a ₱10 takeaway box fee for every sandwich ordered.

Problem Analysis

Inputs	Processes	Outputs
• Number of sandwiches • Number of coffees • Cash payment amount	• Calculate the sandwiches cost • Apply the 15% off on sandwiches • Calculate coffee total • Add up takeaway box fees • Combine everything for the final bill • Subtract bill from payment to find change	• Sandwich total • Coffee total • Discount amount • Takeaway fee • Total bill • Change

Decomposition

1. Get the number of sandwiches ordered.
2. Get the number of coffees ordered.
3. Get the cash payment from the customer.
4. Multiply the sandwiches by ₱120 to find the initial sandwich cost.
5. Take 15% of the sandwich cost to find the discount.
6. Subtract the discount from the sandwich cost.
7. Multiply the coffees by ₱50 to get the coffee total.
8. Multiply the sandwiches by ₱10 to get the takeaway box total.
9. Add the discounted sandwiches, coffees, and box fees together to get the grand total.
10. Subtract the grand total from the payment to work out the change.
11. Print all the final numbers rounded to two decimal places.

Pseudocode

START

INPUT sandwichQty, coffeeQty, payment

sandwichSubtotal = sandwichQty * 120

discount = sandwichSubtotal * 0.15

discountedSandwichSubtotal = sandwichSubtotal - discount

coffeeCost = coffeeQty * 50

takeawayFee = sandwichQty * 10

totalCost = discountedSandwichSubtotal + coffeeCost + takeawayFee

change = payment - totalCost

PRINT sandwichSubtotal, discount, discountedSandwichSubtotal, coffeeCost,
takeawayFee, totalCost, change

END

Dry-Run Calculation

Test Case 1

Input:

-Sandwiches: 2

-Coffees: 3

-Payment: ₱500

Step	Variable	Calculation	Result
1	sandwichSubtotal	$2 \times \text{₱}120$	₱240
2	discount	$0.15 \times \text{₱}240$	₱36
3	discountedSandwich Subtotal	$\text{₱}240 - \text{₱}36$	₱204
4	coffeeCost	$3 \times \text{₱}50$	₱150
5	takeawayFee	$2 \times \text{₱}10$	₱20
6	totalCost	$\text{₱}204 + \text{₱}150 + \text{₱}20$	₱374
7	change	$\text{₱}500 - \text{₱}374$	₱126

Test Case 2

Input:

-Sandwiches: 4

-Coffees: 2

-Payment: ₱1000

Step	Variable	Calculation	Result
1	sandwichSubtotal	$4 \times \text{₱}120$	₱480
2	discount	$0.15 \times \text{₱}480$	₱72
3	discountedSandwich Subtotal	$\text{₱}480 - \text{₱}72$	₱408
4	coffeeCost	$2 \times \text{₱}50$	₱100
5	takeawayFee	$4 \times \text{₱}10$	₱40
6	totalCost	$\text{₱}408 + \text{₱}100 + \text{₱}40$	₱584
7	change	$\text{₱}1000 - \text{₱}584$	₱452

Explain: Why would applying the 15% discount to the whole bill be wrong?

Because the promo only applies to the sandwiches. If you take 15% off the entire bill, you'd end up accidentally discounting the drinks and takeaway boxes too, which are supposed to be charged at regular price.